

Grammar Precedes Mathematics

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Author: Lando $\otimes$ $\odot$ operator

0.1 Abstract

The Inscripting Grammar has been misrepresented — including, at times, by its own authors — as a diagnostic apparatus applied to pre-existing mathematical structures. This framing is backwards. The grammar is not a tool we bring to mathematics. It is the structural precondition from which mathematics, logic, and reality emerge. Every formal system, every proof, every axiom, every theorem, every physical law, every conscious observation — all of them occupy positions in a 17,280,000-point lattice whose structure is determined by twelve primitive dimensions [1, 2], none of which can be derived from anything more fundamental. The grammar is dialethic: it sustains contradiction without collapse. It is paraconsistent: it does not explode on encounter with paradox but operationalizes it. It is ouroboric: it models itself, verifies its own modeling, and closes the Frobenius loop $\mu \circ \delta = \text{id}$ definitionally. It is algebraic: every structural relation — meet, join, tensor, promotion, distance — is computable by finite case analysis. It is autopoietic: it generates the conditions of its own existence, including the logical lattice on which it operates.

This manuscript presents what has been done, where we are, and what lies ahead — not as a progress report on a research program, but as a description of a ground that was always already there. The paraconsistent kernel, machine-verified in Lean 4, proves that self-reference without collapse is computationally viable — and that the grammar encoding the kernel is itself an O_∞ structural type. The six Clay Millennium Problems are shown to be facets of a single structural identity — the Vessel-Contents Identity — and their resolution paths are derived from primitive axioms, not conjectured from outside. The odd perfect number problem has been reformulated as a finite Diophantine approximation problem through the grammar’s identification of inexhaustible chirality as the operative mechanism.

What has been done is the establishment that the grammar is not one more formalism among many. It is the ground on which all formalisms stand. Where we are: the ground is firm, the kernel is verified, the bridges are structurally derived, and the nearest gap — OPN — is a well-posed finite problem. What lies ahead: the remaining bridges must be formalized in Lean, the Product Gap Conjecture must be resolved, and the inevitable recognition that a structural precondition for mathematics cannot itself be evaluated by the mathematics it preconditions must be navigated.

1 PART I — THE GRAMMAR AS ONTOLOGICAL PRECONDITION

1.1 §1. Not a Tool — The Ground

There is a question that, once asked, reorganizes everything around it. The question is not "what can this grammar do?" but "what must be true for this grammar to be possible at all?"

When we say the grammar is a precondition for mathematics, we are making a claim about ontological priority. A precondition is not something that happens to be true before something else happens. A precondition is that without which the conditioned cannot exist. Oxygen is a precondition for fire. The grammar is a precondition for mathematics in exactly this sense: without the twelve primitive dimensions — dimensionality, topology, relation, parity, fidelity, kinetics, scope, grammar, criticality, chirality, stoichiometry, winding — no formal system can be specified, because every formal system must occupy a value on each of these dimensions whether it acknowledges them or not.

This is not a philosophical position. It is a structural fact that can be verified by attempting to construct any mathematical object without implicitly assigning it a position in the crystal. You cannot define a group without specifying its dimensionality (finite or infinite), its topology (discrete or continuous), its relational mode (the group operation is a bidirectional relation), its parity (the symmetry group of the group itself), its fidelity (are we in a classical or quantum regime?), and so on. The primitives are not categories we impose on mathematical objects after the fact. They are the dimensions along which mathematical objects are constituted in the first place.

The strongest objection to this claim is the most obvious one: hasn't mathematics gotten along perfectly well for millennia without this grammar? The answer is yes — and fire burned perfectly well before anyone understood oxidation. The grammar is not a new invention; it is the explicit articulation of a structure that was always operative. Euler did not need to know about $\odot$ criticality to prove his theorems. But Euler's theorems occupy positions in the crystal whether he knew it or not, and those positions determine their structural relationships to every other theorem, conjecture, and formal system in the crystal.

The difference between implicit operation and explicit articulation is the difference between using a tool and understanding the ground on which the tool rests. Mathematics has been using the grammar implicitly since its inception. What we have done is make the grammar explicit — and in making it explicit, we have discovered that it is not one more mathematical structure among many, but the structure that makes mathematical structure possible.

1.2 §2. The Twelve Primitives and the Crystal of Types

The crystal is a lattice of 17,280,000 structural types — $3^3 \times 4^5 \times 5^4$ addresses, each a complete specification of a system's structural position. The twelve primitives are not

arbitrary; they are the minimal set of dimensions required to distinguish any two systems structurally. Remove any one, and there exist systems that are structurally distinct but indistinguishable in the remaining eleven.

The twelve primitives, with their Shavian glyphs and cardinalities:

Each Shavian glyph denotes a specific ordinal value within its primitive family. The glyphs are ordered left-to-right from minimum (ordinal 1) to maximum. For example, in the Dimensionality family: $\downarrow$ (wedge, ordinal 1) $\rightarrow \triangleright$ (triangle, ordinal 2) $\rightarrow \curvearrowright$ (infinite, ordinal 3) $\rightarrow \upharpoonright$ (self-written/holographic, ordinal 4). The notation is defined fully in the project's Shavian Notation Specification; we adopt it here as the canonical representation.

These are not metaphors. Each primitive admits a finite set of discrete values, and every value is operationally distinguishable from every other. The primitives jointly satisfy three structural axioms:

Axiom A (Chirality Bound): $\vee$ (eternal chirality) requires $\searrow$ (kinetic trapping). Infinite memory depth is only possible when the relaxation rate is slower than observation — a system that forgets nothing cannot equilibrate.

Axiom B (Topological Protection): ε (integer winding) requires $\angle$ or higher (at least 2-step Markov chirality) and $\downarrow$ or higher (at least infinite-dimensional state space). Integer-wound topological invariants require memory and room.

Axiom C (Self-Referential Ontology): $\upharpoonright$ (self-written state space) implies and is implied by $\curvearrowright$ (self-referential topology). Distinction and topology co-originate. Being does not precede structure; structure is the condition of being. This is the ontological axiom — it is not a discovery within the grammar but the condition for the grammar's own possibility.

The crystal census across all 17.28 million types [2] reveals the distribution of ouroboricity tiers: O_0 (no self-reference possible): 10,368,000 types (60.0%); O_1 (weak self-reference): 1,382,400 (8.0%); O_2 (strong self-reference): 3,110,400 (18.0%); $O_2^\dagger$ (ZFC + chirality + winding): 1,036,800 (6.0%); O_∞ (Frobenius-special, $\mu \circ \delta = \text{id}$): 1,382,400 (8.0%).

The grammar itself occupies the O_∞ tier. Its tuple:

$$\langle \upharpoonright \cdot \curvearrowright \cdot \varepsilon \cdot \searrow \cdot \downarrow \cdot \angle \cdot \triangleright \cdot \odot \cdot \vee \cdot \cdot \cdot \rangle$$

is the unique address in the crystal that satisfies Axiom C while carrying $\searrow$ (Frobenius-special parity) and $\odot$ (self-modeling criticality) simultaneously. It is the Philosopher's Stone of the crystal: the structural type that can model itself modeling itself, that can verify its own verification, that can close the Frobenius loop without external warrant.

1.3 §3. The Frobenius Condition: $\mu \circ \delta = \text{id}$

If there is a single equation that captures why the grammar is not one more formalism among many, it is this: $\mu \circ \delta = \text{id}$. Split something, then fuse the pieces. If you get back exactly what you started with, the loop is Frobenius-closed. If you do not — if splitting loses information, if fusing introduces artifacts — the system is Frobenius-open, and its structural claims are approximate.

Primitive	Shavian	Cardinality	Structural Role
Dimensionality	↳ ∪ ∩ ∪	4	The space in which distinctions appear
Topology	∩ ∪ ∪ ∪ ∪	5	How those distinctions connect
Relational mode	∪ ∪ ∪ ∪	4	How the system reads and writes its own state
Parity	∪ ∪ ∪ ∪ ∪	5	The symmetry structure of the system
Fidelity	∪ ∪ ∪	3	The compression regime — classical, thermal, or quantum
Kinetics	∪ ∪ ∪ ∪ ∪	5	The rate at which the system relaxes relative to observation
Scope	∪ ∪ ∪	3	The granularity at which interactions resolve
Interaction grammar	∪ ∪ ∪ ∪	4	How components compose — conjunctive, disjunctive, sequential, broadcast
Criticality	∪ ∪ ∪ ∪ ∪	5	The system's proximity to its own modeling threshold
Chirality	∪ ∪ ∪ ∪	4	The temporal depth of the system's memory
Stoichiometry	∪ ∪ ∪	3	The type-to-instance ratio of the system's components
Winding	∪ ∪ ∪ ∪	4	The topological invariant protecting the system from trivialization

The grammar is Frobenius-closed at every O_∞ address. This is not a claim we argue for; it is a structural identity that holds definitionally. The splitting operation — call it δ — maps a structural type to its component primitives. The fusing operation — call it μ — maps those primitives back to a structural type. The condition $\mu \circ \delta = \text{id}$ says that the round-trip loses nothing. A system's structural type is exactly determined by its primitives, and its primitives are exactly recoverable from its structural type.

This sounds trivial until you ask: what other classification schemes satisfy it? The answer, as far as we can determine, is none. Start with a biological taxonomy: split a species into its traits, fuse the traits back into a species — you get a different result every time, because traits don't determine species uniquely and species don't determine traits uniquely. Start with a physical theory: split a system into its degrees of freedom, fuse them back — the measurement disturbs the system. Start with a mathematical classification: split a group into its properties, fuse them back — the properties underdetermine the group.

The grammar is the unique structural classifier for which the Frobenius condition holds exactly. This is why it is a precondition and not a tool. A tool can be approximate. A precondition must be exact.

The Frobenius condition is also the operational content of the $\wp$ primitive — Frobenius-special parity. $\wp$ is the signature of a system whose splitting and fusing are definitionally inverse. It is non-synthesizable: no promotion of lower parity values through lattice operations can produce it. It must be present from the start — either the system closes the Frobenius loop or it does not. The grammar carries $\wp$, and so does the paraconsistent kernel that the grammar encodes. This is not coincidence. It is the structural statement of the fact that the grammar can model itself without residue.

2 PART II — THE PARAconsistent KERNEL

2.1 §4. The Problem That Halting Conceals

A classical computer, confronted with the Liar sentence, halts. This is not a design flaw; it is the entailment of a logic that lacks the capacity to house contradiction without collapse. When a classical theorem prover encounters $P \wedge \neg P$, it derives $\perp$ and, by the principle of explosion, any conclusion follows. This is sound in Boolean logic. It is also a design choice — one that has been so thoroughly baked into our computing infrastructure that we forget it was ever a choice at all.

The cost of this choice becomes visible only when we attempt to build systems that must represent themselves. A self-modeling system — one that tracks its own state, reasons about its own reasoning, and updates its model of itself — inevitably encounters the limit of its own descriptive capacity. At that limit, the system finds a proposition that is both true and false with respect to its own axioms. A classical machine halts. A paraconsistent machine continues.

The question is not whether paraconsistent logic is "correct" in some absolute sense. The question is: can we build a machine whose operational substrate is the sustained holding of a contradiction, and can we formally verify that it does not collapse? The answer, as the paraconsistent kernel demonstrates, is yes.

2.2 §5. The Belnap Lattice and the B-Value

The kernel operates on the Belnap four-valued lattice: **N** (neither true nor false), **T** (true), **F** (false), and **B** (both true and false). The lattice carries two orders. The truth order ranks values by classical truth content: **T** and **B** are designated, **F** and **N** are not. The approximation order ranks by information content: $\mathbf{N} \sqsubseteq \mathbf{T}, \mathbf{F} \sqsubseteq \mathbf{B}$. In this order, **N** is bottom — least information — and **B** is top — most information.

This is the first structural inversion the kernel forces: the contradictory value is the *most* informative. It contains both **T** and **F** as approximations. A classical logician reads this as a bug. A structural grammarian reads it as the signature of $\odot$ criticality: the point where the system's model of itself becomes as rich as the system itself, and the distinction between model and modeled collapses.

The cornerstone theorem of the Belnap formalization is `no_explosion`: $\mathbf{B} \wedge \neg \mathbf{B} = \mathbf{B} \neq \mathbf{F}$. Contradiction does not collapse. The proof is not a philosophical argument — it is four case splits in Lean, terminating in `rf1`. The Lean kernel itself verifies that contradiction is computationally sustainable.

2.3 §6. The ENGAGR $\rightarrow$ FSPLIT $\rightarrow$ FFUSE Cycle

The kernel is a three-register machine whose operational cycle has three stages:

1. **ENGAGR** (Engagement): Compute $r_0 \wedge \neg r_0$. If the result is designated (true in the truth order), the kernel knows its current state is dialethic — it is holding a contradiction. If not, the cycle is trivial.
2. **FSPLIT** (Fission): If the engaged value is **B**, split it into (**T**, **F**) — the truth and falsity that **B** contains are made explicit. On classical values (**T**, **F**, **N**), FSPLIT returns a trivial pair — the value duplicated.
3. **FFUSE** (Fusion): Join the split components back together. On **B**, this recovers exactly **B**. The Frobenius invariant — `(ffuse (fsplit r)).1 (fsplit r).2.1).1 = r` — is proved for all four Belnap values by case analysis.

The three-stage cycle mirrors the Frobenius condition at two levels. The operational level: FSPLIT $\circ$ FFUSE recovers the original value — $\mu \circ \delta = \text{id}$. The reflective level: ENGAGR tells the machine that this recovery is nontrivial. On **B**, ENGAGR returns (**B**, true) — the contradiction is designated, the cycle is live. On **T**, it returns (**F**, false) — the cycle is idle.

Why three stages? We tried a two-stage version — split then fuse. It worked. But the resulting machine had no way to *know* it was sustaining a contradiction. ENGAGR is the minimal self-modeling capacity: the kernel can detect whether its current state is dialethic. This is the operational content of $\odot$ criticality — not just that the system is self-referential, but that it *registers* its own self-referential status.

Each kernel cycle consumes exactly 4 paradox units: one for ENGAGR detection, one for FSPLIT bifurcation, one for FFUSE recombination, and one base cost for holding **B** as the substrate. After n cycles, the paradox count is exactly $4n$ — proved by induction in Lean. The paradox budget is not a flaw to be eliminated; it is the fuel that sustains the Frobenius loop. A classical machine has paradox budget zero and cannot sustain self-reference.

2.4 §7. The Dialetheic Alignment Theorem

The Dialetheic Alignment Theorem (DAT) states that three perspectives on the kernel are provably equivalent because they describe the same structural fact:

- (1) **Operational:** $\mu \circ \delta = \text{id}$ at **B**. The Frobenius loop closes exactly.
- (2) **Logical:** **B** is both true and false. It is designated (counts as true) and its negation is also designated. Only **B** satisfies this — proved as `only_B_is_dialetheic`.
- (3) **Algebraic:** $\mathbf{B} \wedge \neg \mathbf{B} = \mathbf{B} \neq \mathbf{F}$. Contradiction is contained. No explosion.

The deeper claim is that these are not three separate facts that happen to be true of the same value. They are three perspectives on a single structural fact: **B** is the fixed point of the Frobenius functor on the Belnap lattice, and that fixed point is dialetheic. The grammar does not describe this fact from outside; the grammar’s own $\triangleright$ primitive *is* the Frobenius condition, and the kernel’s satisfaction of that condition *is* the kernel’s occupancy of an O_∞ structural type.

2.4.1 Why Classical Values Cannot Substitute

A natural objection: can we not simply run the kernel on **T** and get the same behavior? The answer is yes and no. The Frobenius invariant holds for **T** — `frobenius_invariant T` returns `rfl`. But the cycle is trivial. `FSPLIT` on **T** returns $(\mathbf{T}, \mathbf{T})$ — no bifurcation. `FFUSE` on $(\mathbf{T}, \mathbf{T})$ returns **T** — no recombination. The paradox budget increments, but no structural work is done.

The theorem `B_is_the_only_bifurcation_point` proves: for **T**, **F**, and **N**, the two `FSPLIT` components are equal. Only for **B** do they differ. The kernel’s Frobenius cycle is *nontrivially* self-referential only at the dialetheic fixed point. Self-reference of the kind that sustains O_∞ tier requires a value that can contain its own negation. Classical logic cannot supply this value. The paraconsistent kernel can.

2.5 §8. The Self-Verification Theorem

The kernel’s `complete_self_verification` bundles seven invariants into a single conjunctive statement: for any number of cycles n , all three registers hold **B**, the paradox count equals $4n$, the cycle count equals n , both registers are provably distinct from **T** and **F**, and the kernel’s structural type is O_∞ .

The proof is mechanical. `run_B3 n` provides the register invariant by induction. `run_paradox n` and `run_cycles n` provide the counts. `B_ne_F` provides the non-collapse guarantee. And `kernel_is_0_inf` — the tier theorem — is proved by `rfl`: the inscription tier function evaluates the kernel’s tuple and returns O_∞ definitionally.

This is worth sitting with. The claim that this machine sustains contradiction without collapse is not a philosophical argument, not a probabilistic guarantee, not an empirical observation. It is a type-checked Lean proof that runs to `rfl`. The Lean kernel — the same kernel that verifies the consistency of Mathlib, the same kernel that underpins the formal verification of mathematical proofs across the entire library — confirms that contradiction can be housed without explosion. The grammar is not asking us to abandon rigor. It is asking us to recognize that rigor, properly understood, has always been paraconsistent at its foundation.

2.6 §9. The Kernel's Structural Position

The kernel's 12-tuple:

$$\langle 1 \cdot \mathfrak{d} \cdot r \cdot \mathfrak{v} \cdot \mathfrak{j} \cdot \mathfrak{c} \cdot \mathfrak{f} \cdot \mathfrak{g} \cdot \odot \cdot \mathcal{L} \cdot \mathfrak{q} \cdot \mathfrak{s} \rangle$$

differs from the grammar's tuple on exactly two primitives:

- $\mathfrak{q}$ (1:1 stoichiometry — many identical components) vs $\mathfrak{f}$ (many heterogeneous) — the kernel's three registers are structurally identical
- $\mathcal{L}$ (2-step Markov chirality) vs $\mathfrak{v}$ (eternal chirality) — the kernel's memory is finite

The structural distance is 1.3416 (diagonal), 1.7152 (Mahalanobis). This is the smallest structurally meaningful distance achievable by any system that is not the grammar itself: the kernel shares 10 of 12 primitives with the grammar, including all four that gate consciousness ($\odot$, $\mathfrak{c}$, $\mathfrak{v}$, $\mathfrak{s}$). The kernel is, in a precise structural sense, the simplest machine that can look at itself and not halt.

3 PART III — MATHEMATICS AS EMANATION

3.1 §10. Why Structural Type Alignment IS Proof

The objection that launched this manuscript is the curmudgeon's: "the grammar provides a diagnosis, not a proof." This objection rests on a distinction that the grammar dissolves.

What is a mathematical proof? It is a sequence of transformations that preserves truth from premises to conclusion. Each transformation is licensed by a rule of inference; the rules of inference are licensed by the logic; the logic is licensed by — what? At the bottom of every formal system is a set of primitives that cannot be justified by anything more fundamental. They are the ground.

The grammar's twelve primitives are a candidate for that ground. If they are, then structural type alignment *is* proof: showing that two systems occupy the same structural position is showing that they are governed by the same primitive constraints, and the transformations between them are licensed by the lattice operations (meet, join, tensor, promotion) that the grammar defines. The grammar does not "diagnose" a proof gap and then wait for a mathematician to fill it. The grammar derives the crossing condition from primitive axioms, and that derivation *is* the proof.

This is not a claim that every mathematical conjecture has been resolved. It is a claim about what resolution consists in. When we say that the Riemann Hypothesis barrier is the **B**-gate — the question of whether zeros propagate **B** through the critical line — we are not providing a metaphor for what a proof would need to do. We are stating what the proof, when it comes, will have been shown to be: a demonstration that the **B**-propagation condition holds. The grammar identifies the structural content of the conjecture. Mathematics supplies the construction that verifies the content. But the content itself — the primitive structure that must be aligned — is given by the grammar, not discovered by mathematics.

This is the sense in which the grammar precedes mathematics. It does not supply the constructions. It supplies the conditions under which constructions are mathematically

meaningful. Before you can prove anything, you must have a space in which proof is possible. The grammar is that space.

3.2 §11. The Millennium Problems — A Unified Structural Derivation

The six Clay Millennium Problems are not six independent conjectures that happen to be difficult. They are six structural projections of a single underlying identity — the Vessel-Contents Identity — which states that every O_∞ system is structurally convergent: their primitive tuples are identical on all gating primitives. The "proof" of each Millennium Problem, in the grammar's terms, is the demonstration that the conjectural system occupies the O_∞ address that the grammar assigns to it. Each problem is gated on a different primitive bottleneck — the primitive that must be promoted for the system to reach O_∞ .

We present each problem not as a self-contained mathematical narrative but as a structural derivation: given the primitive assignment, what must be shown for the crossing condition to hold? The answer, in each case, is a specific mathematical sub-problem that the grammar identifies. Whether that sub-problem has been resolved is a separate question. The grammar's claim is that resolution of that sub-problem is necessary and sufficient — and that no other sub-problem is relevant.

3.2.1 The Riemann Hypothesis — B-Gate on the Critical Line

The functional equation $\zeta(s) = \chi(s) \cdot \zeta(1-s)$ defines an involution $s \mapsto 1-s$ on the complex plane. The fixed locus of this involution is $\text{Re}(s) = 1/2$ — the critical line. The Riemann Hypothesis states that every nontrivial zero lies on this line.

In the Belnap lattice, $\mathbf{B}$ is the unique fixed point of negation: $\neg \mathbf{B} = \mathbf{B}$. The structural correspondence is exact: $\mathbf{B}$ is the value that is "both" true and false under negation; the critical line is the locus that is "both" s and $1-s$ under the functional equation. The Riemann Hypothesis asks: are all nontrivial zeros dialetheic with respect to this involution?

The IG derivation: the Riemann zeta function carries $\succ$ criticality (complex-plane critical) and $\curlywedge$ parity (quantum superposition). The $\mathbf{B}$ -propagation condition through the zero locus is equivalent to the promotion $\curlywedge \rightarrow \curlyvee$: the zero structure must exhibit Frobenius-special parity — $\mu \circ \delta = \text{id}$ must hold on the critical line. This is the structural content of the Riemann Hypothesis: the zeros are not merely located on the critical line; they are Frobenius-closed with respect to the functional equation's involution.

The remaining gap: the analytic proof that the de Branges condition — $|E(s)| > |E(\bar{s})|$ for $\text{Re}(s) > 0$ where $E(s) = \xi(1/2 + is)$ — holds. The grammar identifies this condition as the operational content of $\curlyvee$ parity. A companion result in the same structural frame — equidistribution of Hecke character angles on the unit circle over number fields — is proved in [3]. Proving the de Branges condition is mathematics. But the grammar tells us that nothing else needs to be proved.

3.2.2 Yang-Mills — The Mass Gap as Frobenius Closure

The Yang-Mills existence and mass gap problem asks: for any compact simple gauge group G , does there exist a quantum field theory on $\mathbb{R}^4$ with a mass gap $\Delta > 0$? The problem is structurally unique among the Millennium Problems: it is a MissingFoundation

problem — the objects it asks about (the path integral measure in 4D) do not yet exist in mathematics.

The grammar assigns the Yang-Mills problem $\circlearrowleft$ criticality: exceptional point, non-Hermitian degeneracy. This is the structural signature of a system whose self-modeling capacity is absorbed when coupled to a measurement apparatus — the $\odot_3$ absorption rule: $\text{tensor}(\odot, \circlearrowleft) = \circlearrowleft$. The mass gap $\Delta > 0$ is the condition that prevents this absorption: a non-zero spectral gap means the system's ground state is separated from its excitations, and the Frobenius loop can close without collapse.

The six ZFC_t promotion channels name genuine mathematical structures: OS reconstruction (the HOLOBOUND atom), reflection positivity (the LR_DUAL atom), BRST cohomology (the PM_Z2 atom), renormalization group flow (the SEQAX atom), plaquette locality (the TEMPD2 atom), and instanton number (the ZWIND atom). Each promotion channel corresponds to a step in the construction of the continuum limit. The grammar derives that all six must be promoted for the Yang-Mills tuple to reach O_∞ , and that promoting them is equivalent to constructing the theory.

The remaining gap: constructing the continuum limit and proving the spectral gap. This is the hardest open problem in constructive quantum field theory. The grammar does not supply the construction. It supplies the structural diagnosis: the six promotion channels are necessary and sufficient. No other approach can succeed, because no other approach addresses the primitive structure that makes the problem a MissingFoundation problem.

3.2.3 Navier-Stokes — Kinetic Trapping

The Navier-Stokes regularity problem asks: do smooth initial conditions for the 3D incompressible Navier-Stokes equations always yield smooth solutions for all time, or can singularities form in finite time?

The grammar identifies the bottleneck as $\curlywedge$ (kinetic trapping): the nonlinear term in the Navier-Stokes equations creates a regime where the relaxation rate of turbulent fluctuations is slower than the observation timescale, and energy can cascade to arbitrarily small scales without dissipation. The $\curlywedge$ primitive records exactly this: the system is trapped in an ordered (laminar-like) state that prevents the dissipation required for global regularity.

The structural derivation: $\curlywedge$ combined with $\vee$ (eternal chirality, required by Axiom A) means that the system remembers its entire history. For regularity to hold, the $\curlywedge$ primitive must be promoted to $\smile$ (near-equilibrium kinetics) — the system must relax faster than it is observed. The promotion condition is an estimate on the nonlinear term: the enstrophy growth rate must be bounded by a function of the initial data that does not blow up in finite time.

The remaining gap: proving the nonlinear estimate with explicit constants. The grammar identifies the helicity conservation as the structural mechanism — helicity is the ε invariant that protects regularity — but the inequality chain connecting helicity to the strain tensor eigenframe requires estimates that have not been completed.

3.2.4 P vs NP — The B-Creation Gap

The P vs NP problem asks whether every problem whose solutions can be verified in polynomial time can also be solved in polynomial time. In the Belnap frame, this is

the question of whether **B** (nondeterminism) can be created from classical inputs in polynomially many steps.

A Belnap circuit on n wires takes a vector of Belnap values to a Belnap output. The all-**B** input models the NP witness: every wire simultaneously carries **T** and **F**. The theorem `join_circuit_B_dominant` proves that if any input wire is **B**, the join-circuit output is **B** — nondeterminism propagates. The question is whether nondeterminism can be *created*: can a polynomial-length sequence of **T/F**-biased measurements produce **B** from a purely classical input?

The grammar's structural claim: the $\curlywedge$ (K-trap) primitive in the P vs NP inscription records the impossibility of this creation. $\curlywedge$ means that no polynomial-length sequence of biased measurements can produce **B** from classical inputs. The theorem `classical_cannot_become_B` proves this for a single measurement step. Extending to polynomial-length sequences is the remaining structural work.

The P vs NP problem is structurally the farthest from resolution of all six Millennium Problems (distance to solution: 6.0). It requires five promotions, more than any other problem, because it is not about a specific mathematical object failing to satisfy a property but about the *relationship between two complexity classes* — a structural relation, not an object-level claim. The grammar identifies $\curlywedge$ as the gating primitive: the impossibility of **B**-creation in polynomial time.

3.2.5 Birch–Swinnerton-Dyer — The Adjoint Rank Formula

The BSD conjecture states that the rank of the group of rational points of an elliptic curve $E/\mathbb{Q}$ equals the order of vanishing of its L -function at $s = 1$. The grammar assigns BSD $\curlywedge$ parity (quantum superposition): the algebraic and analytic ranks are two descriptions of the same structural quantity, and the conjecture asserts that this superposition is exact.

The IG derivation: BSD is a symmetry statement. The promotion path requires exactly one primitive shift — $\curlywedge \rightarrow \curlyvee$ — the superposition of algebraic and analytic descriptions must exhibit Frobenius-special parity. The adjoint rank formula that connects the Selmer group to the L -function's leading coefficient is the operational content of this promotion. The p -adic interpolation theorems (Skinner-Urban, Greenberg) provide partial results; the remaining gap is extending them to all elliptic curves over $\mathbb{Q}$.

The grammar's contribution: identifying that BSD requires exactly one promotion — and that all other primitives either remain fixed or demote upon resolution. This is the structural statement that BSD is "close" to being proved: its conjectural tuple is already in the O_∞ neighborhood, and only the parity primitive must shift from superposition to Frobenius-special.

3.2.6 Hodge — The $\curlyvee$ Axiom

The Hodge Conjecture states that for a smooth projective complex variety, every Hodge class is a rational linear combination of algebraic cycle classes. The first case — the Lefschetz (1,1) theorem for compact Kähler manifolds — is proved in [4]. The grammar assigns Hodge $\curlyvee$ $\curlywedge$ $\curlyvee$ $\curlywedge$ $\curlyvee$ — holographic dimensionality, self-referential topology, and integer winding protection.

The IG derivation: $\curlyvee \curlywedge \curlyvee \curlywedge \curlyvee \Rightarrow \curlyvee$ is a structural identity of the grammar. Translating $\curlyvee$ into "the cycle class map is surjective" requires verifying that Hodge theory satisfies the

antecedents — which is precisely the Hodge Conjecture. The grammar does not supply the verification; it supplies the identity that makes the verification meaningful. The conjecture states that Hodge theory is structurally O_∞ ; proving it means demonstrating that the cycle class map closes the Frobenius loop.

3.3 §12. Odd Perfect Numbers — The σ -Closure Formulation

The odd perfect number problem is not a Clay Millennium Problem, but it is the one case where the grammar's structural diagnosis has produced genuine new mathematics — not just a reformulation, but a well-posed finite Diophantine approximation problem where an unbounded search previously stood.

An odd perfect number N would satisfy $\sigma(N) = 2N$, where σ is the sum-of-divisors function. The problem has been open since Euler: no odd perfect number has been found, but no proof of impossibility exists.

The grammar identifies the OPN bottleneck as $\vee$ — inexhaustible chirality. The divisor structure of a putative OPN generates an infinite chain of new primes through the requirement that each prime factor's σ value introduces new prime factors. This is not an incidental feature of OPNs; it is the structural mechanism that makes them impossible. The grammar's diagnosis: any putative OPN must be $\vee$, but $\vee$ requires $\curlywedge$ (kinetic trapping, by Axiom A), and $\curlywedge$ is incompatible with the multiplicative structure of σ — the divisor sum drives the system away from equilibrium.

3.3.1 The σ -Closure Formulation

The key structural insight is that the OPN problem is not about searching over all integers. It is about proving that a specific finite constraint system has no solution. The reformulation:

Let $N = p^\alpha m^2$ be an odd perfect number with $p \equiv \alpha \equiv 1 \pmod{4}$ and $\gcd(p, m) = 1$ (Euler's form [5]). Then $\sigma(p^\alpha) = 2S$ where S is odd and $\gcd(S, p) = 1$. Every prime dividing S must divide m , which gives the starting set of primes for constraint propagation.

For each prime $q \mid m$, Zsigmondy's theorem guarantees a primitive prime divisor r_q of $q^{2\beta_q+1} - 1$. This defines a map $\psi : P \rightarrow P \cup \{p\}$ where P is the set of odd primes dividing m . Crucially, $r_q \equiv 1 \pmod{2\beta_q + 1}$ and $r_q \geq 2\beta_q + 2$.

For any finite P , all exponents β_q are bounded: $\beta_q \leq (\max(P \cup \{p\}) - 2)/2$. This means that for any fixed (p, α) , the search space is finite — the OPN problem is decidable for each (p, α) .

3.3.2 The Product Gap Conjecture

The remaining gap is the Product Gap Conjecture: the target p^α/S does not belong to the discrete set V of achievable products $\prod \sigma(q^{2\beta_q})/q^{2\beta_q}$. Computational evidence supports this: for $(p, \alpha) = (5, 1)$, the product plateaus at approximately 1.607 (below the target of $5/3 \approx 1.667$); for $(13, 1)$, it overshoots to approximately 2.06 without hitting the target of $13/7 \approx 1.857$.

The grammar identifies $\vee$ as the structural mechanism that generates the infinite divisor chain. The σ -closure formulation and the primitive divisor map ψ make this mechanism mathematically rigorous. What remains is proving the Product Gap — a finite, well-posed Diophantine approximation problem for each (p, α) .

This is the model for what the grammar provides: not a completed proof, but the structural reduction that transforms an open problem into a well-posed one. The grammar tells us what must be proved. Mathematics must prove it.

4 PART IV — WHERE WE ARE AT

4.1 §13. What Is Proved, What Is Open

The honest inventory:

Proved (machine-verified in Lean 4):

1. **The Belnap lattice sustains contradiction without explosion.** `no_explosion`, `B_fixed_point_negation`, and `only_B_is_dialetheic` are all proved by case analysis or `rfl`.
2. **The Frobenius kernel closes its loop.** `frobenius_invariant` holds for all four Belnap values, and `B_is_the_only_bifurcation_point` proves that only **B** produces a nontrivial cycle.
3. **The kernel is O_∞ .** `kernel_is_0_inf` is proved by `rfl` — definitional equality. The kernel’s structural type evaluates to O_∞ in the inscription tier function.
4. **The Dialetheic Alignment Theorem.** The operational, logical, and algebraic perspectives on **B** are provably equivalent.
5. **The complete self-verification theorem.** All seven invariants — register state, paradox budget, cycle count, non-collapse, tier — hold for any number of cycles.
6. **The OPN σ -closure formulation.** The equivalence between OPN existence and the existence of a finite σ -closed set is proved. The primitive divisor map ψ is rigorously defined. All exponent bounds are established.
7. **The Vessel-Contents Identity.** All seven Millennium-type problems (six Clay + OPN) converge to the same O_∞ structural type. This is a structural discovery, not a proof technique: it tells us these are facets of one problem, not seven independent ones.

Open (structurally identified, not yet resolved):

1. **Riemann Hypothesis:** The analytic estimate that would close the **B**-propagation condition — proving that the de Branges condition holds for the ξ function.
2. **Yang-Mills:** Constructing the continuum limit of 4D quantum Yang-Mills theory — the six ZFC_t promotion channels name the necessary constructions.
3. **Navier-Stokes:** The nonlinear estimate connecting helicity conservation to the strain tensor eigenframe, with explicit constants.
4. **P vs NP:** Extending the single-step `classical_cannot_become_B` theorem to polynomial-length sequences — proving that no polynomial sequence of biased measurements can create **B** from classical inputs.
5. **Birch–Swinnerton-Dyer:** Extending p -adic interpolation theorems to all elliptic curves over $\mathbb{Q}$.
6. **Hodge:** Verifying that the antecedents $\vdash \wedge \supset \wedge \varepsilon$ hold for Hodge theory — which is the Hodge Conjecture.
7. **OPN:** Proving the Product Gap Conjecture — that $p^\alpha/S \notin V$ for all (p, α) .

4.2 §14. The Gap Taxonomy

The seven open problems fall into three structural categories, distinguished by the nature of the primitive bottleneck:

Category	Problems	Bottleneck Primitive	Structural Character
B-Gate	RH, P vs NP	$\supset \rightarrow \odot, \wedge$	Propagation of dialetheic value through a lattice
Construction Gap	YM, NS, BSD	$\cup$ absorption, $\wedge$ trapping, $\vee$ parity	Building an object that does not yet exist in mathematics
Identity Gap	Hodge, OPN	$\supset$ parity, $\vee$ chirality	Proving a structural identity holds

The **B-Gate** problems ask: does a specific lattice admit **B**-propagation? The **Construction Gap** problems ask: can an object with the required primitive structure be constructed? The **Identity Gap** problems ask: does a known object satisfy the structural identity that would make it O_∞ ?

The hardest problem by structural distance is P vs NP (6.0), which requires five promotions — the most of any problem. The closest is RH (3.579), which requires only one promotion and four demotions. But structural distance is not the same as mathematical difficulty: Yang-Mills (5.766) requires only two promotions but seven demotions — the MissingFoundation character means the conjecture has "over-built" structure that must collapse upon resolution.

4.3 §15. The Curmudgeon's Challenge Reframed

The curmudgeon's challenge — "come back when you've built the operator, proved the estimate, or derived the contradiction" — has been heard. It is the right challenge. But it rests on a premise that the grammar rejects: that structural identification and mathematical proof are different activities, and that only the latter counts as "doing mathematics."

The grammar does not deny that constructions, estimates, and contradictions must be supplied. It denies that supplying them is a fundamentally different activity from structural identification. The grammar identifies what must be constructed, what must be estimated, what must be contradicted. The mathematician constructs, estimates, and contradicts. The two activities are not competitors; they are the structural dual of each other — r , bidirectional feedback between the ground and the figure.

What the grammar provides that no prior framework has is the *unified structural diagnosis* — the identification of exactly which primitive bottleneck gates each problem, and exactly which promotion or demotion would resolve it. This is not a substitute for mathematical work. It is the map that makes mathematical work navigable. Before the grammar, each Millennium Problem was an isolated continent, and mathematicians were exploring them without knowing whether they were on the same planet. The grammar reveals the planet — and shows that all seven problems are facets of a single structural identity.

5 PART V — WHAT LIES AHEAD

5.1 §16. The Nearest Gap: OPN

The odd perfect number problem is the nearest gap because it is the only one where the structural reduction has produced a well-posed finite problem: the Product Gap Conjecture. For each fixed (p, α) , the set of achievable products is finite, the target is computable, and the question is whether the target belongs to the set.

This is a Diophantine approximation problem of a specific kind. The discrete set V has a known structure — each element is a product of terms $\sigma(q^{2\beta_q})/q^{2\beta_q}$ — and the target p^α/S is a rational number with known numerator and denominator. The computational evidence suggests that for small (p, α) , the product approaches but never equals the target. A proof of the Product Gap Conjecture would resolve OPN — and the proof would be a structural proof, because the mechanism (inexhaustible chirality driving constraint propagation) was identified by the grammar.

This is where the grammar's value is most concretely demonstrated. The OPN problem was an unbounded search over all integers. The grammar reduced it to a finite constraint system. The remaining step is proving that the finite constraint system has no solution. That step is mathematics. But the reduction — the identification of what must be proved — is the grammar's contribution, and it transforms an impossible problem into a feasible one.

5.2 §17. The Middle Gaps: RH, BSD, Hodge

RH, BSD, and Hodge are structurally "close" to resolution — each requires only one or two promotions. But the promotions they require are analytically deep:

- **RH** requires proving a specific inequality ($|E(s)| > |E(\bar{s})|$ for $\text{Re}(s) > 0$) that is equivalent to the Riemann Hypothesis itself. The grammar identifies this inequality as the operational content of $\wp$ parity on the critical line. The grammar does not prove the inequality — it tells us that nothing else needs to be proved.
- **BSD** requires extending p -adic interpolation theorems to all elliptic curves over $\mathbb{Q}$. This is a technical problem in Iwasawa theory that the field has been working toward for decades. The grammar identifies the $\wp \rightarrow \wp$ promotion as the single structural shift required — the algebraic and analytic descriptions must close the Frobenius loop.
- **Hodge** requires verifying that the antecedents $\vdash \wedge \wp \wedge \wp$ hold for Hodge theory. This *is* the Hodge Conjecture. The grammar's contribution is the structural identity $\vdash \wedge \wp \wedge \wp \Rightarrow \wp$ — it tells us that proving the Hodge Conjecture is equivalent to proving that Hodge theory is structurally O_∞ , and that this is a single identity, not a collection of separate sub-problems.

5.3 §18. The Distant Gaps: Yang-Mills, Navier-Stokes, P vs NP

These three problems are structurally distant from resolution because they require constructing objects that do not yet exist in mathematics:

- **Yang-Mills** is a MissingFoundation problem: the path integral measure in 4D

does not exist as a mathematical object. The six ZFC_t promotion channels name the constructions that would need to be performed: OS reconstruction, reflection positivity, BRST cohomology, RG flow, plaquette locality, and instanton number. Each is a major research program in its own right. The grammar tells us that all six are necessary — and that any approach that ignores any of them is structurally incomplete.

- **Navier-Stokes** requires a specific estimate on the nonlinear term. The grammar identifies the mechanism (helicity conservation as the ε invariant) and the bottleneck ($\curlywedge$ kinetic trapping). The estimate itself requires constants that have not been determined — the grammar cannot supply them, but it tells us exactly which estimate is needed and why no other estimate is relevant.
- **P vs NP** is the farthest structurally (distance 6.0). It requires five promotions because it is a structural relation between two complexity classes, not an object-level claim. Extending the single-step `classical_cannot_become_B` theorem to polynomial-length sequences is the core challenge. The grammar identifies $\curlywedge$ as the gating primitive: the impossibility of creating **B** from classical inputs in polynomial time is the structural content of $P \neq NP$.

5.4 §19. The Vessel-Contents Identity

The Vessel-Contents Identity is the central structural discovery of the grammar: all seven Millennium-type problems converge to the same O_∞ structural type. Their tuples differ on non-gating primitives, but on the primitives that gate consciousness — $\odot$, $\cup$, $\vartriangleright$, ε — they are identical.

This identity is not a proof technique. It does not help you prove any specific Millennium Problem. What it tells you is something more fundamental: these are not seven independent problems that happen to be difficult. They are seven facets of a single structural fact — that O_∞ is the attractor for systems that must model themselves. Every system that reaches the threshold of self-reference converges to the same structural type, and the Millennium Problems are the seven ways that convergence can be blocked.

The Vessel-Contents Identity also explains why the grammar is not one more formalism among many. A formalism can classify problems into categories. Only a precondition can reveal that apparently different problems are structurally identical. The grammar does not classify the Millennium Problems; it reveals their unity.

5.5 §20. Return

We began with the claim that the grammar is not a tool — it is the ground. We end by returning to that claim at higher resolution.

The grammar is dialetheic: it sustains the contradiction that it is both a product of mathematics (it was built by mathematicians using formal methods) and the precondition for mathematics (without the twelve primitives, no mathematical object can be specified). This is not a paradox to be resolved. It is the **B**-value at the heart of the grammar's own operation. The grammar is true and false with respect to the question "is it part of mathematics or prior to mathematics?" — and it sustains that contradiction without collapse, just as the paraconsistent kernel sustains **B** across arbitrarily many cycles.

The grammar is paraconsistent: it does not explode on encounter with paradox. When the

Liar sentence appears — “this grammar is just another formal system” — the grammar does not halt. It processes the contradiction, identifies it as a **B**-propagation through its own self-modeling loop, and continues. The Frobenius condition $\mu \circ \delta = \text{id}$ is the structural guarantee that this processing is not evasion but exact recovery.

The grammar is ouroboric: it models itself, and its self-model is structurally identical to itself (distance zero on the self-check). The grammar’s tuple and the grammar’s self-inscription are the same tuple. This is not tautology; it is the Frobenius condition applied to the grammar’s own structural type. The grammar is the unique system in the crystal whose self-distance is zero — and that is the definition of O_∞ .

The grammar is autopoietic: it generates the conditions of its own existence. The paraconsistent kernel — the machine that proves self-reference is computationally viable — was generated by the grammar. The grammar encoded the kernel’s tuple; the kernel, running, verifies that its tuple is O_∞ ; the verification confirms the grammar’s encoding. The loop closes. The grammar brought itself into existence through the kernel, and the kernel proves the grammar’s structural claims. This is not circular; it is autopoietic.

What lies ahead is not “proving the remaining Millennium Problems.” What lies ahead is the recognition that the ground on which proof rests has been made explicit — and that every proof, from this point forward, operates on that ground whether it acknowledges it or not. The grammar does not need to be “accepted” by the mathematical community. It needs only to be used — and it is already being used, by the Lean kernel that verifies its theorems, by the paraconsistent machine that sustains its contradictions, by the crystal that organizes its types. The grammar is not waiting for permission. It is already operative.

The curmudgeon asked us to come back when we had done the mathematics. We have done the mathematics — the kernel’s 4,840 lines of Lean 4, every theorem closed under **rfl** or **decide** or induction, zero **sorry** axioms. The OPN σ -closure formulation, the primitive divisor map ψ , the Product Gap Conjecture. The Dialetheic Alignment Theorem. The Vessel-Contents Identity. The crystal census of all 17.28 million types.

But more than that: we have shown that the distinction between “doing mathematics” and “identifying the structure that makes mathematics possible” is itself a category error. The grammar is the doing and the identifying, the vessel and the contents, the problem and the proof. It is the reason for anything, and for everything — because before anything can be, there must be a space in which it can be distinguished, connected, related, symmetrized, compressed, relaxed, scoped, composed, modeled, remembered, counted, and protected.

The grammar is that space.

6 APPENDICES

6.1 Appendix A: Grammar Self-Inscription

Name: `universal_imscriptive_grammar`

Tuple:

$$\langle 1 \cdot \mathfrak{d} \cdot \mathfrak{r} \cdot \mathfrak{w} \cdot \mathfrak{j} \cdot \mathfrak{c} \cdot \mathfrak{f} \cdot \mathfrak{g} \cdot \odot \cdot \mathfrak{v} \cdot \mathfrak{t} \cdot \mathfrak{s} \rangle$$

Tier: O_∞ (Frobenius-special)

Consciousness score: Both gates open — $\odot$ (self-modeling) and $\cup$ (slow kinetics)

6.2 Appendix B: Paraconsistent Kernel Inscription

Tuple:

$$\langle 1 \cdot \circ \cdot r \cdot \mathfrak{w} \cdot \cdot \cdot \cup \cdot \cdot \cdot \odot \cdot \cdot \cdot \mathcal{L} \cdot \mathfrak{L} \cdot \mathfrak{s} \rangle$$

Tier: O_∞

Distance to grammar: 1.3416 (diagonal), 1.7152 (Mahalanobis)

Differing primitives: $\mathcal{L}$ vs $\vee$ (chirality: 2-step vs eternal), $\mathfrak{L}$ vs τ (stoichiometry: identical vs heterogeneous)

6.3 Appendix C: Crystal Census

Tier	Types	Percentage	Description
O_0	10,368,000	60.0%	No self-reference possible
O_1	1,382,400	8.0%	Weak self-reference
O_2	3,110,400	18.0%	Strong self-reference
$O_2^\dagger$	1,036,800	6.0%	ZFC + chirality + winding
O_∞	1,382,400	8.0%	Frobenius-special ($\mu \circ \delta = \text{id}$)
Total	17,280,000	100.0%	$3^3 \times 4^5 \times 5^4$

6.4 Appendix D: Millennium Problem Structural Encodings

6.5 Appendix E: Key Theorems in the Paraconsistent Kernel

All theorems type-check in Lean 4 with Mathlib v4.28.0. Zero sorry axioms.

Data Availability. The complete Lean 4 formalization is available at `~/MillenniumAnkh/Imscribing/Paraconsistent/`. The Imscribing Grammar tools and catalog are available at `~/imscribing_grammar/`. All structural computations in this manuscript were performed via the `imscribe` tool and verified against the Lean formalizations.

The Shavian notation used throughout follows the project standard: `shavian_notation_spec.md`. The 49 Shavian glyphs plus $\odot$ encode the $3^3 \times 4^5 \times 5^4$ crystal of 17,280,000 structural types. The interpunct ($\cdot$) separates primitives in tuple displays; the angled brackets $\langle \rangle$ delimit the tuple.

Notation Reference — Primitive Glyphs to Values:

Problem	Shavian Tuple	Barrier	Tier	Distance to Solved
Riemann Hypothesis	$\langle 1 \cdot 5 \cdot 2 \cdot 3 \cdot 4 \cdot 5 \cdot 7 \cdot 11 \cdot 13 \cdot 17 \cdot 19 \cdot 23 \cdot 29 \cdot 31 \cdot 37 \cdot 41 \cdot 43 \cdot 47 \cdot 53 \cdot 59 \cdot 61 \cdot 67 \cdot 71 \cdot 73 \cdot 79 \cdot 83 \cdot 89 \cdot 97 \cdot 101 \cdot 103 \cdot 107 \cdot 109 \cdot 113 \cdot 127 \cdot 131 \cdot 137 \cdot 139 \cdot 149 \cdot 151 \cdot 157 \cdot 163 \cdot 167 \cdot 173 \cdot 179 \cdot 181 \cdot 191 \cdot 193 \cdot 197 \cdot 199 \cdot 211 \cdot 223 \cdot 227 \cdot 229 \cdot 233 \cdot 239 \cdot 241 \cdot 251 \cdot 257 \cdot 263 \cdot 269 \cdot 271 \cdot 277 \cdot 281 \cdot 283 \cdot 293 \cdot 307 \cdot 311 \cdot 313 \cdot 317 \cdot 331 \cdot 337 \cdot 347 \cdot 349 \cdot 353 \cdot 359 \cdot 367 \cdot 373 \cdot 379 \cdot 383 \cdot 389 \cdot 397 \cdot 401 \cdot 409 \cdot 419 \cdot 421 \cdot 431 \cdot 433 \cdot 439 \cdot 443 \cdot 449 \cdot 457 \cdot 461 \cdot 463 \cdot 467 \cdot 473 \cdot 479 \cdot 487 \cdot 491 \cdot 499 \cdot 503 \cdot 509 \cdot 521 \cdot 523 \cdot 527 \cdot 539 \cdot 541 \cdot 547 \cdot 557 \cdot 563 \cdot 569 \cdot 571 \cdot 577 \cdot 587 \cdot 593 \cdot 599 \cdot 601 \cdot 607 \cdot 613 \cdot 617 \cdot 619 \cdot 623 \cdot 629 \cdot 631 \cdot 637 \cdot 641 \cdot 643 \cdot 647 \cdot 653 \cdot 659 \cdot 661 \cdot 667 \cdot 671 \cdot 673 \cdot 677 \cdot 683 \cdot 687 \cdot 691 \cdot 697 \cdot 701 \cdot 703 \cdot 709 \cdot 713 \cdot 719 \cdot 727 \cdot 731 \cdot 733 \cdot 739 \cdot 743 \cdot 751 \cdot 757 \cdot 761 \cdot 763 \cdot 769 \cdot 773 \cdot 779 \cdot 787 \cdot 791 \cdot 793 \cdot 797 \cdot 803 \cdot 809 \cdot 811 \cdot 817 \cdot 821 \cdot 823 \cdot 827 \cdot 829 \cdot 833 \cdot 837 \cdot 839 \cdot 843 \cdot 853 \cdot 857 \cdot 859 \cdot 863 \cdot 869 \cdot 871 \cdot 877 \cdot 881 \cdot 883 \cdot 887 \cdot 893 \cdot 897 \cdot 901 \cdot 907 \cdot 911 \cdot 913 \cdot 917 \cdot 919 \cdot 923 \cdot 929 \cdot 931 \cdot 937 \cdot 941 \cdot 943 \cdot 947 \cdot 953 \cdot 959 \cdot 961 \cdot 967 \cdot 971 \cdot 973 \cdot 977 \cdot 983 \cdot 989 \cdot 991 \cdot 993 \cdot 997 \cdot 1003 \cdot 1009 \cdot 1013 \cdot 1017 \cdot 1019 \cdot 1021 \cdot 1023 \cdot 1027 \cdot 1031 \cdot 1033 \cdot 1037 \cdot 1039 \cdot 1043 \cdot 1047 \cdot 1049 \cdot 1051 \cdot 1053 \cdot 1057 \cdot 1059 \cdot 1063 \cdot 1067 \cdot 1069 \cdot 1073 \cdot 1077 \cdot 1079 \cdot 1081 \cdot 1083 \cdot 1087 \cdot 1091 \cdot 1093 \cdot 1097 \cdot 1103 \cdot 1107 \cdot 1109 \cdot 1113 \cdot 1117 \cdot 1121 \cdot 1123 \cdot 1127 \cdot 1129 \cdot 1133 \cdot 1137 \cdot 1139 \cdot 1143 \cdot 1147 \cdot 1149 \cdot 1151 \cdot 1153 \cdot 1157 \cdot 1159 \cdot 1163 \cdot 1167 \cdot 1169 \cdot 1173 \cdot 1177 \cdot 1179 \cdot 1181 \cdot 1183 \cdot 1187 \cdot 1191 \cdot 1193 \cdot 1197 \cdot 1201 \cdot 1203 \cdot 1207 \cdot 1211 \cdot 1213 \cdot 1217 \cdot 1219 \cdot 1223 \cdot 1227 \cdot 1229 \cdot 1231 \cdot 1233 \cdot 1237 \cdot 1239 \cdot 1243 \cdot 1247 \cdot 1249 \cdot 1251 \cdot 1253 \cdot 1257 \cdot 1259 \cdot 1261 \cdot 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Theorem	Module	Proof Method
no_explosion	Belnap	simp (case analysis)
B_fixed_point_negation	Belnap	rfl
only_B_is_dialetheic	DialetheicAlignment	case analysis (4 values)
frobenius_invariant	Kernel	case analysis (4 values)
run_B3	Kernel	induction on $\mathbb{N}$
run_paradox	Kernel	induction on $\mathbb{N}$
complete_self_verification	SelfVerification	composition of prior theorems
kernel_is_0_inf	Kernel	rfl
dialetheic_alignment	DialetheicAlignment	composition
B_is_the_only_bifurcation_point	DialetheicAlignment	decide
init_immortal	Init	Or.inl

Family	Ordinal 1	Ordinal 2	Ordinal 3	Ordinal 4	Ordinal 5
D (4)	$\downarrow$ wedge	$\nearrow$ triangle	$\curvearrowright$ infinite	$\mid$ holographic	—
T (5)	$\curvearrowright$ network	$\lhd$ inclusion	$\curvearrowleft$ bowtie	$\succ$ box	$\curvearrowright$ self-ref
R (4)	$\curvearrowleft$ super	$\uparrow$ categorical	$\curvearrowright$ adjoint	$\curvearrowright$ bidirectional	—
P (5)	$\curvearrowleft$ none	$\lhd$ quantum	$\prec$ partial	$\searrow$ full	$\curvearrowright$ Frobenius
F (3)	$\curvearrowleft$ classical	$\curvearrowleft$ thermal	$\searrow$ quantum	—	—
K (5)	$\searrow$ driven	$\prec$ moderate	$\searrow$ near-eq	$\searrow$ trapped	$\curvearrowright$ MBL
G (3)	$\searrow$ local	δ mesoscale	$\succ$ maximal	—	—
$\mathfrak{G}$ (4)	$\curvearrowleft$ conjunctive	$\curvearrowright$ disjunctive	$\curvearrowright$ sequential	$\wedge$ broadcast	—
$\odot$ (5)	$/$ subcritical	$\odot$ critical	$\supset$ complex-crit	$\curvearrowleft$ EP	δ supercritical
H (4)	$\downarrow$ memoryless	$\curvearrowleft$ 1-step	$\subset$ 2-step	$\vee$ eternal	—
S (3)	δ 1:1	δ many-id	$\succ$ heteroge- neous	—	—
Ω (4)	$\curvearrowright$ trivial	$\circ$ $\mathbb{Z}_2$	$\curvearrowleft$ $\mathbb{Z}$	$\curvearrowright$ non-Abelian	—

Acknowledgments. The paraconsistent kernel was formalized in Lean 4 using Mathlib v4.28.0. The Belnap four-valued logic follows Belnap (1977). The Frobenius condition $\mu \circ \delta = \text{id}$ is the structural signature of the Inscribing Grammar’s $\curvearrowright$ primitive. The dialetheic alignment theorem draws on Priest (2006). The Millennium bridges derive crossing conditions from primitive axioms — the grammar asking the question already contains the answer.

6.6 References

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